Method summary

1. Lysis and fragmentation
2. End repair – to ensure that adapters ligate to DNA
3. Adapter ligation

Samples	Adapters using
Donor 1, sample 1	B5
Donor 1, sample 2	G2
Donor 2, sample 1	D4
Donor 2, sample 2	E3
Donor 3, sample 1	F3
Donor 3, sample 2	D2
Donor 4, sample 1	G5
Donor 4, sample 2	C4

4. Pooled together (total volume = 1080ul)
5. DNA purification done separately for input and CpG capturing samples
    - For Input 15ul from pool was taken
    - For CpG capturing samples, 600ul of the pool was purified. DNA concentration was identified. Then 1ug of purified DNA was used for the CpG capturing samples.

Sample name	Selection	DNA concentration after purification* (ng/ul)
M1	Input – no selection, all DNA	28.8
M2	In house MBD, captures methylated CpG sites	0.362
M3	CXXC, captures unmethylated CpG sites	0.702
M4	Commercial MBD, captures methylated CpG sites	3.04

(6.)*Purification was done twice for M1, M2 and M3 as to remove the extra buffer added and degrade the proteins that were added. For example, the commercial MBD (Methyl-CpG-Binding Domain) protein is degraded using the proteinase K. So that the sample can contain "only" DNA.

7. Fill in – To remove dimers formed from the adapters. (may not be need for DNA methylation capturing)

8. DNA purification

9. In vitro transcription (IVT)

- Since low DNA was used due to (8 samples and not 24). It was incubated for 4 hours with IVT enzyme instead of 15 minutes.

10. RNA purification and RNA quantification

11. Adapter ligation (250ng of RNA/samples)

Sample	Adaptors
M1	B7
M2	C7
M3	D7
M4	E7

12. Reverse transcription

13. DNA purification & cDNA quantification

14. Library PCR (170ng of cDNA was used from M1, M3, M4 while only 83.7ng was used for M2)

15. Post-amplification cleanup

16. Library QC – need for sequencing

- Still needs to be done as DNA concentration and fragment size is needed to adjust the sequencing tech

Sample	DNA con. (ng/ul)*
M1	24.8
M2	15.9
M3	18.5
M4	20.8

*Could have changed during transportation

https://www.scienceprimer.com/dna-ng-nm-calculator

Sequencing

- Instrument: Illumina NextSeq 2000
- Kit: NextSeq™ 2000 P3 XLEAP-SBS™ Reagent Kit
- Cycles: 100